

# Junyoung Park

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## RESEARCH INTERESTS

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- Distributional data analysis and optimal transport, with applications to wearable device data
- Dimension reduction of compositional data, with applications to microbiome data
- Non-Euclidean data analysis

## PROFESSIONAL POSITIONS

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### Postdoctoral Research Fellow

09/2024 –

Department of Biostatistics, University of Michigan, MI, USA  
Supervisor: Irina Gaynanova

### BK21 Postdoctoral Research Fellow

03/2024 – 08/2024

Natural Science Research Institute, KAIST, Daejeon, Korea  
Supervisor: Cheolwoo Park

## EDUCATION

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### Ph.D. in Mathematical Sciences

03/2018 – 02/2024

KAIST, Daejeon, Korea  
Thesis: “Kernel Methods for Compositional Data and Dimensionality Reduction”  
(Co)Advisors: Cheolwoo Park, Jeongyoun Ahn

### B.S. in Mathematics

03/2013 – 02/2018

Korea University, Seoul, Korea

## AWARDS AND SCHOLARSHIPS

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### *Competitive Travel Awards*

- **IMS New Researchers Conference**, Boston, MA, USA 2026
- **IMS New Researchers Travel Award** 2026
- **Long Program on Connectomics**, Institute of Mathematical and Statistical Innovation 2026
- **1st IMS International New Researchers Conference**, Seville, Spain 2025

### *Awards and Scholarships in Korea*

- **Best Ph.D. Dissertation Award**, Korean Statistical Society, 2 awardees/year; prize  $\approx$  \$2,000 2025
- **Presentation Award for Graduate Students**, Korean Statistical Society; prize  $\approx$  \$200 2022
- **Outstanding Teaching Assistant Award**, Calculus II, KAIST Fall, 2020
- **University Students Contest of Mathematics**, Silver Awards, Korean Mathematical Society 2016, 2017
- **Presidential Science Undergraduate Fellowship**, fully funded for 8 semesters 2013–2018
- **The Korean Mathematical Olympiad (KMO)** Gold Awards in 2nd round, middle school division 2009

## EXPERIENCE

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### Research Assistant in Statistical Learning

KAIST

Under the supervision of Prof. Cheolwoo Park and Jeongyoun Ahn

08/2021 – 02/2024

– Kernel methods, compositional data, dimension reduction

### Research Assistant in Algebraic Geometry

KAIST

Under the supervision of Prof. Sijong Kwak

09/2018 – 07/2021

– Projective algebraic geometry, syzygies, applied algebraic geometry

### Technical Research Personnel

KAIST

For military obligation in South Korea

03/2020 – 02/2023

## RESEARCH PAPERS

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Underline denotes mentored Ph.D. student

*Preprints/Submitted:*

9. **Park, J.** and Gaynanova, I. (2026+) “Fréchet regression of multivariate distributions with nonparanormal transport.” Submitted. Preprint available on [arXiv:2603.07014](#).  
– 2026 IMS New Researchers Travel Award
8. Shao, E., **Park, J.**, Punjabi, N., Jiang, H, and Gaynanova, I. (2026+) “Fast distance computation between multivariate distributions via nonparanormal Transport.” Preprint available on [arXiv:2603.00322](#). (E. Shao received IMS Hannan Graduate Student Travel Award 2026)
7. Xu, C., Schmidt, B.M., Krambrink, A., **Park, J.**, Song, P.X.-K., Jones, T.L.Z., Holmes, C., Chen, K., Ye, W., Kolenic, G., Roy, S., Sen, C., Pop-Busui, R., Spino, C., and Gaynanova, I. (2026+) “Profiling Longitudinal Wound Healing: Analysis of Healing Rates and the Predictive Power of Wound Area and Duration in the Diabetic Foot Consortium.” Submitted.
6. **Park, J.**, Park, C., and Ahn, J. (2026+) “A geometry-preserving approach to interpretable dimension reduction of compositional data.” Submitted. Preprint available on [arXiv:2509.05563](#).

*Peer-reviewed publications:*

5. **Park, J.**, Kok, N., and Gaynanova, I. (2026+) “Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions.” *Statistics in Medicine*, Accepted. Preprint [link](#). R package [OptiThresholdR](#).
4. Yeon, K., Gao, F., Kim, R., Lee, D., **Park, J.**, Brown, D.A., and Park, C. (2026+) “A Bayesian Approach to Semi-Supervised Domain Adaptation in Streaming Data.” *Journal of Statistical Computation and Simulation*, 96(9), 2237–2262. [Link](#).
3. **Park, J.**, Ahn, J., and Park, C. (2023), “Kernel Sufficient Dimension Reduction and Variable Selection for Compositional Data via Amalgamation.” *International Conference on Machine Learning (ICML)*, pp. 27034–27047, PMLR. [Link](#).
2. Kang, I., Choi, H., Yoon, Y.-J., **Park, J.**, Kwon, S.-S., and Park, C. (2023), “Frechet Distance-Based Cluster Analysis for Multi-Dimensional Functional Data.” *Statistics and Computing*, 33(4), 75. [Link](#).
1. **Park, J.**, Yoon, C., Park, C., and Ahn, J. (2022), “Kernel Methods for Radial Transformed Compositional Data with Many Zeros.” *International Conference on Machine Learning (ICML)*, pp. 17458–17472, PMLR. [Link](#)

*In preparation:*

- Multiple Imputation of Missing Predictors and Inference in Fréchet Regression of Distributional Responses (**Park, J.**, Szeto, N., and Gaynanova, I.).

## TALKS

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### *Invited talks:*

- (Scheduled) **IMSI Long Program**, during the invited visit to Connectomics program 12/2026
- (Scheduled) International Conference on Computational Statistics (CompStat) 08/2026
- **1st IMS International New Researchers Conference** 12/2025
  - Title: Fréchet regression of multivariate distributions under the semiparametric nonparanormal model
- **KAIST, Graduate School of Data Science** 07/2025
  - Title: Distributional data approaches for wearable device data: optimal thresholds and beyond

### *Contributed talks/posters:*

- **Mobile Technologies Research Innovation Collaborative (MeTRIC) Symposium** (poster) 01/2026
  - Title: Beyond fixed thresholds: optimizing summaries of wearable device data
- **IMS International Conference on Statistics and Data Science (ICSDS)** 12/2025
  - Title: Fast Fréchet regression of multivariate distributions with application to wearable device data
- **Joint Statistical Meetings** 08/2025
  - Title: Beyond fixed thresholds: optimizing summaries of wearable device data
- **Joint Statistical Meetings** 08/2024
  - Title: Interpretable dimension reduction for compositional data
- **Winter Conference, the Korean Statistical Society** 12/2023
  - Title: Interpretable composition-to-composition dimension reduction via conditional covariance operator
- **International Conference on Machine Learning (ICML)** (poster) 07/2023
  - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **Summer Conference, the Korean Statistical Society** 06/2023
  - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **KAIST Mathematical Sciences Graduate student Seminar** 10/2022
  - Title: Kernel methods for radial transformed compositional data with many zeros
- **International Conference on Machine Learning (ICML)** (spotlight talk) 06/2022
  - Title: Kernel methods for radial transformed compositional data with many zeros
  - Presented also at 2022 Summer Conference, the Korean Statistical Society, (**presentation award**)

## ACADEMIC SERVICES

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- **Journal Refereeing**
  - Journal of Royal Statistical Society, Series B (1); Journal of American Statistical Association (1); Annals of Applied Statistics (2); Biometrics (3); WIREs Computational Statistics (1)

## TEACHING

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- **Teaching Assistant** at KAIST (selected without duplication)
  - Statistical Data Science Practice (DS516) Spring 2023
  - Probability and Statistics (MAS250) Fall 2021
  - Abstract Algebra I (MAS311) Spring 2021
  - Mathematical Statistics (MAS355) Fall 2019
  - Matrix Group Theory (MAS435) Spring 2019
    - Gave a guest lecture on connectedness of Lie groups (in English)
  - Abstract Algebra II (MAS312) Fall 2018
    - Gave several guest lectures throughout the semester

## COMPUTING SKILLS

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- Python (for machine learning, statistics), R (for statistics), previous experiences with Matlab and C
- Optimization acceleration: with deep learning frameworks (**TensorFlow** and **Pytorch**), with just-in-time compiler (**numba**)
- Linux environment and high-performance computing (HPC) cluster systems

## LANGUAGES

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- English
- Korean

## HOBBIES

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- Singing, better with playing guitar.
- Running, hiking, and climbing.